$$\int_0^t Rt\,dt = \frac{1}{2}Rt^2$$

$$= \frac{1}{2}qt \quad (Rt = q)$$

= Area of the inventory triangle OAP (Fig.6.12)

Cost of holding inventory per production run = $\frac{1}{2}C_1Rt^2$

Set up cost per production run $= C_3$.

Total cost per production run = $\frac{1}{2}C_1Rt^2 + C_3$

Average total cost per unit time:

$$C(t) = \frac{1}{2}C_1Rt + \frac{C_3}{t} \quad \ldots (1)$$

$$\frac{d}{dt}C(t) = \frac{1}{2}C_1R - \frac{C_3}{t^2} \quad \ldots (2)$$

$$\frac{d^2C(t)}{dt^2} = \frac{2C_3}{t^3} \quad \ldots (3)$$

$C(t)$ is minimum if $\frac{d}{dt}C(t) = 0$ and $\frac{d^2}{dt^2}C(t) > 0$

$$\frac{d}{dt}C(t) = 0 \Rightarrow \frac{1}{2}C_1R - \frac{C_3}{t^2} = 0$$

$$\Rightarrow t = \sqrt{\frac{2C_3}{C_1R}}$$

When $\sqrt{\frac{2C_3}{C_1R}}$, $\frac{d^2C(t)}{dt^2} = \frac{2C_3}{\left(\frac{2C_3}{C_1R}\right)^{\frac{3}{2}}} > 0$

Thus $C(t)$ is minimum for optimum time interval $t_0 = \sqrt{\frac{2C_3}{C_1R}}$

Optimum quantity $q_0$ to be produced during each production run,

$$EOQ = q_0 = Rt_0 = \sqrt{\frac{2C_3R}{C_1}}$$

This is known as the Optimal Lot-size formula due to Wilson.

(i) Optimum number of orders per year

$$n_0 = \frac{\text{demand}}{\text{EOQ}} = R\sqrt{\frac{C_1}{2C_3R}} = \sqrt{\frac{RC_1}{2C_3}} = \frac{1}{t_0}$$

(ii) Minimum inventory cost per unit time, $C_0 = \sqrt{2C_1C_3R}$

(iii) Carrying cost $= \frac{q_0}{2} \times C_1$,

Ordering cost $= \frac{R}{q_0} \times C_3$

(iv) At $EOQ$, Ordering cost = Carrying cost

(v) Total optimum cost = Rp + $\sqrt{2C_1C_3R}$

Here, we will discuss $EOQ$ problems only without shortage cost.

**Example 6.30**

A company uses 48000 units/year of a raw material costing ₹ 2.5 per unit. Placing each order costs ₹45 and the carrying cost is 10.8 % per year of the unit price. Find the $EOQ$, total number of orders per year and time between each order. Also verify that at $EOQ$ carrying cost is equal to ordering cost.

***Solution:***

Here demand rate: $R = 48000$

Inventory cost: $C_1 = 10.8\%$ of $2.5 = \frac{10.8}{100} \times 2.5 = 0.27$

Ordering cost: $C_3 = 45$

Economic order quantity: $q_0 = \sqrt{\frac{2C_3R}{C_1}}$

$$= \sqrt{\frac{2 \times 45 \times 48000}{0.27}} = 4000 \text{ units}$$

Number of orders per year = $\frac{R}{q_0} = \frac{48000}{4000} = 12$

Time between orders: $t_0 = \frac{q_0}{R} = \frac{1}{12} = 0.083$ year

At $EOQ$, carrying cost:

$$= \frac{q_0}{2} \times C_1 = \frac{4000}{2} \times 0.27 = ₹\ 540$$

Ordering cost $= \frac{R}{q_0} \times C_3 = \frac{48000}{4000} \times 45 = ₹\ 540$

So at $EOQ$ carrying cost is equal to ordering cost.



06_11th_BM-STAT_Ch-6-EM.indd 147 21-04-2020 12:22:23 PM